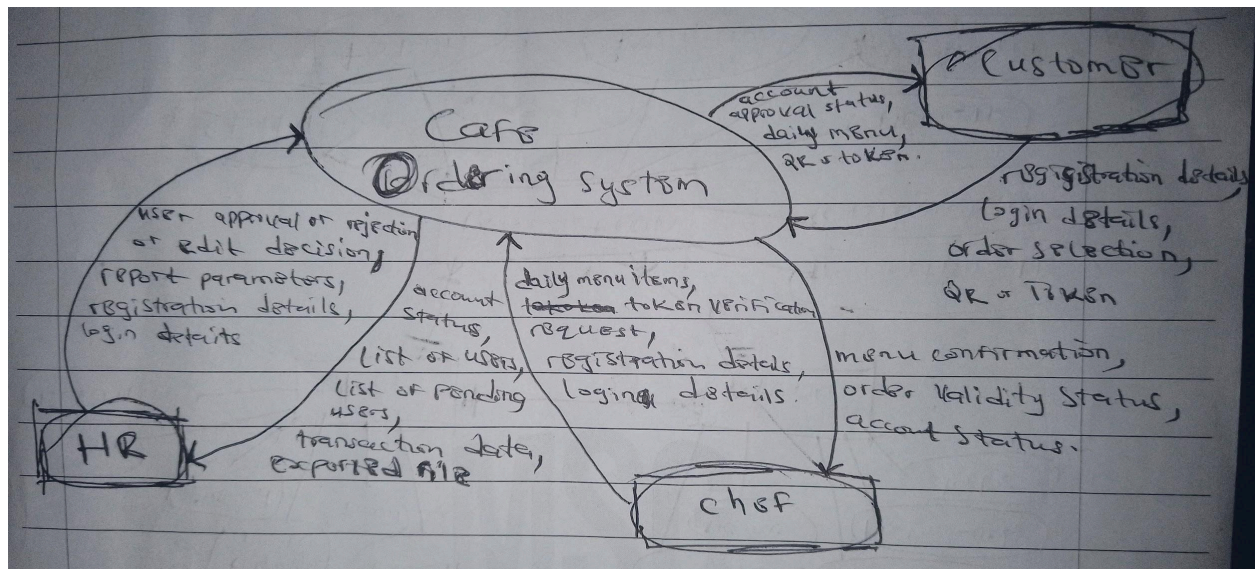


Cafe Ordering System

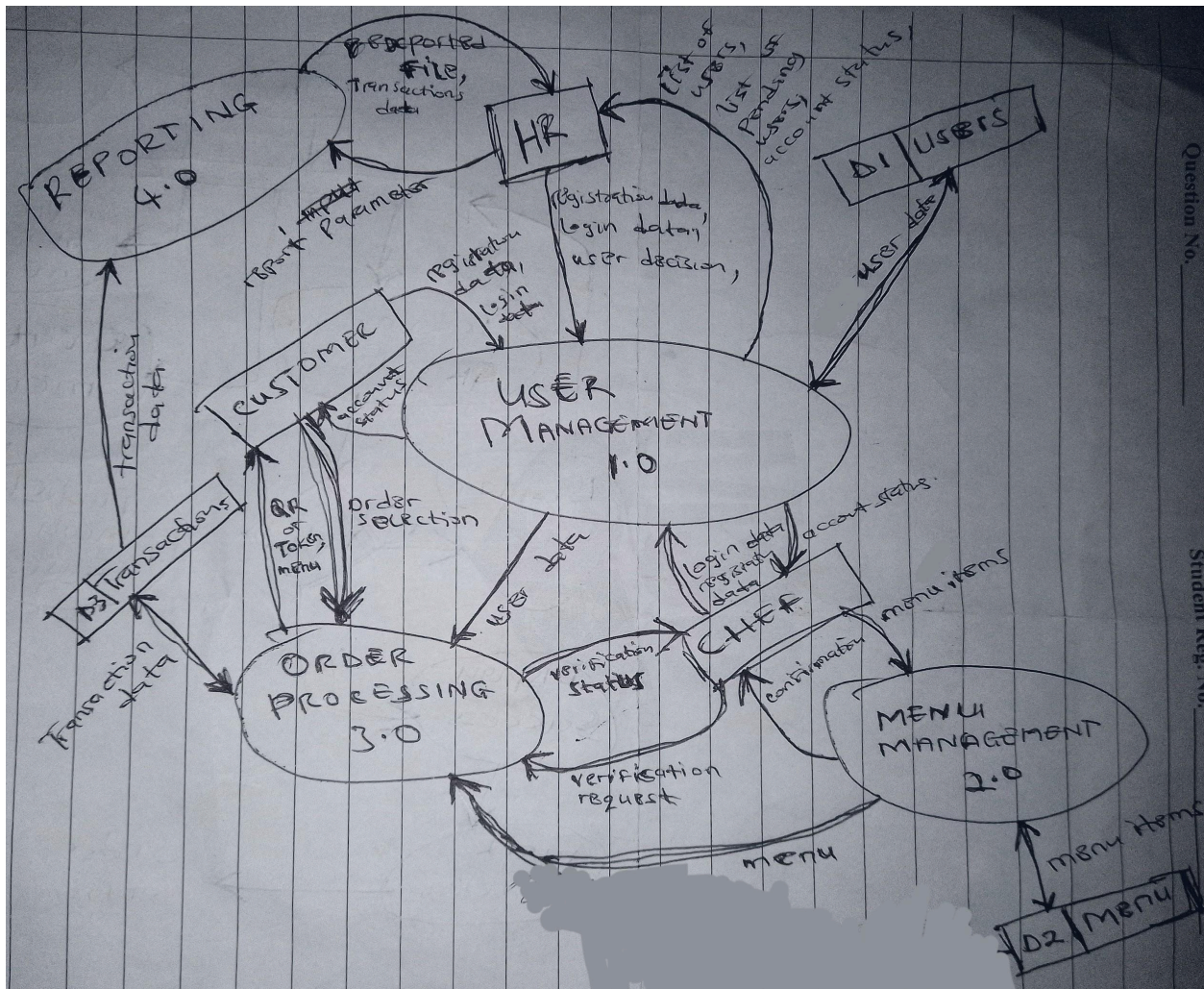
DFDs and Processes

Sketches provide a much clearer roadmap for the actual data dependencies within the system. By declaring handshakes between sub processes, the logic for the backend implementation becomes airtight.

Here is a preview of **DFD Level 0** and **Level 1** diagrams.



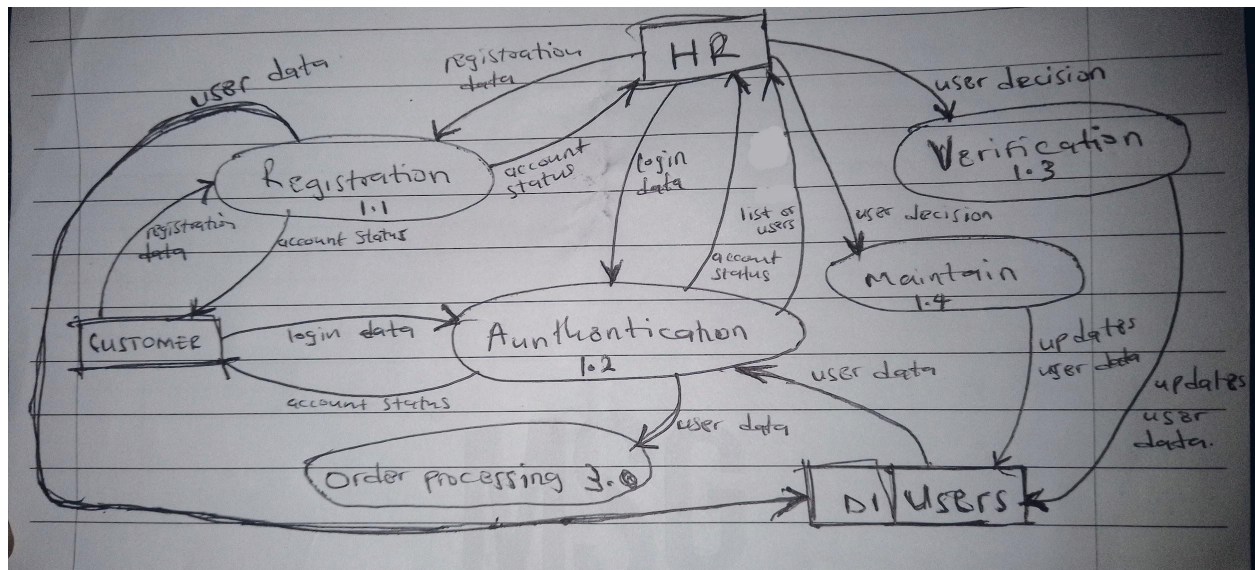
Level 0



Level 1

Here is the finalized technical breakdown for the **DFD Level 2** based on requirements and diagrams.

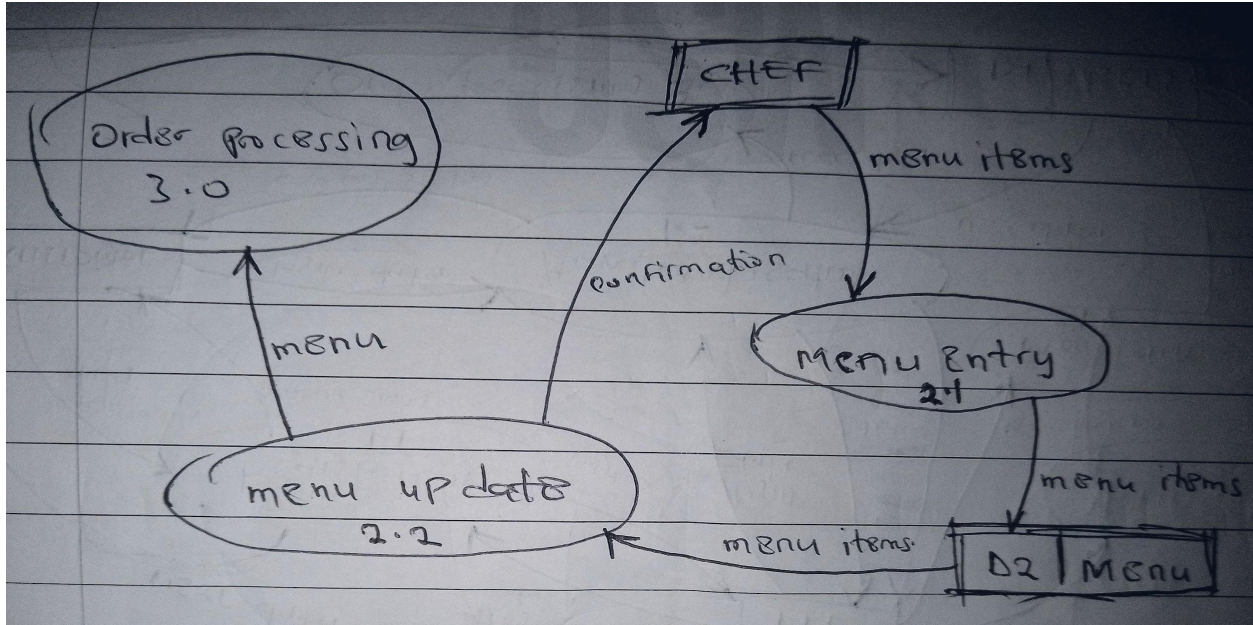
1.0 User Management



This module serves as the primary gateway, handling registration and identity verification for all system actors.

- **1.1 Registration:** Receives **Registration Data** from all actors (Customer, Chef, HR). It logs this **User Data** directly into **D1: Users** and returns an **Account Status** (Success/Failure) to the respective actor.
- **1.2 Authentication:** Receives **Login Data** from any actor and returns an **Account Status** upon authentication success or failure.
 - **External Integration:** For **Process 3.2 (Transaction)**, it sends **Authenticated User Data** to verify customers and include their details in records.
 - **Administrative Access:** Provides the **HR Manager** with a **List of Users** (both verified and unverified) for oversight.
- **1.3 Verification:** Handles the **User Decision** from the HR Manager to update a user's verification status in **D1: Users**.
- **1.4 Maintain:** Manages ongoing **User Decisions** from HR to update or edit user records in **D1: Users**.

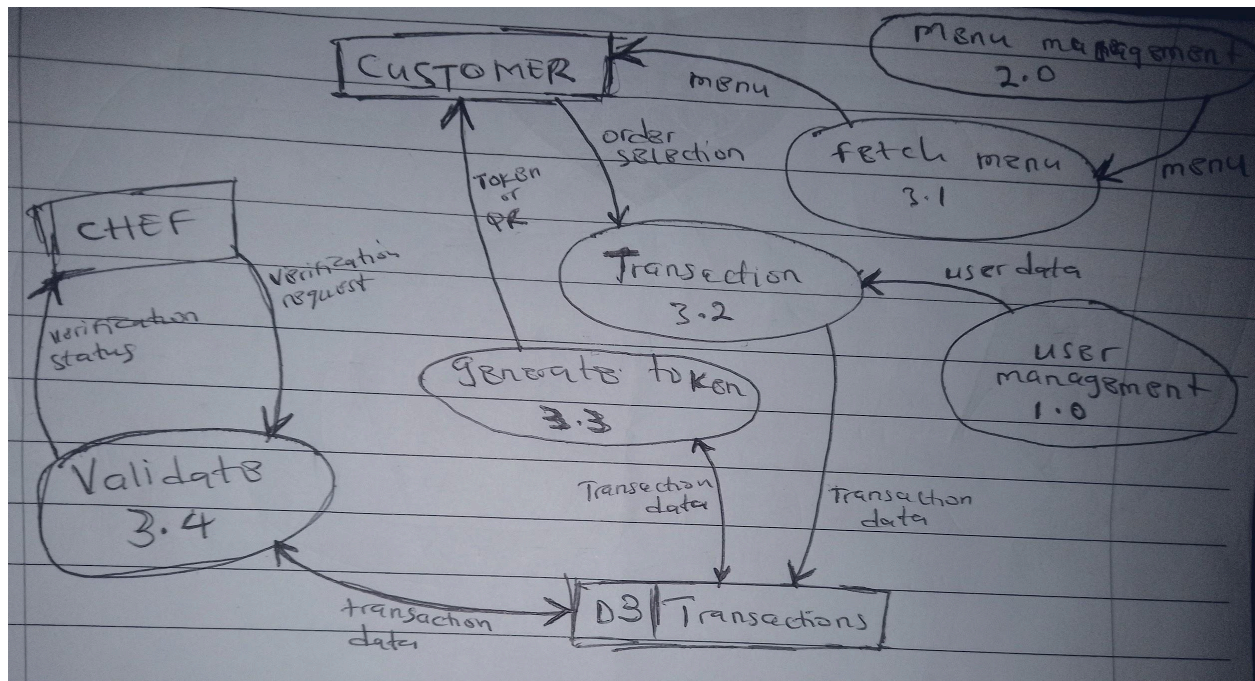
2.0 Menu Management



This module centralizes menu control, ensuring the kitchen can update availability in real-time.

- **2.1 Menu Entry:** The **Chef** inputs **Menu Items**, which are stored in **D2: Menu**.
- **2.2 Menu Update:** Pulls current items from **D2: Menu** to provide a **Confirmation** to the **Chef** after a successful entry.
 - **Downstream Flow:** It is responsible for sending **Menu Data** to **Process 3.1 (Fetch Menu)** so customers can see the current offerings before ordering.

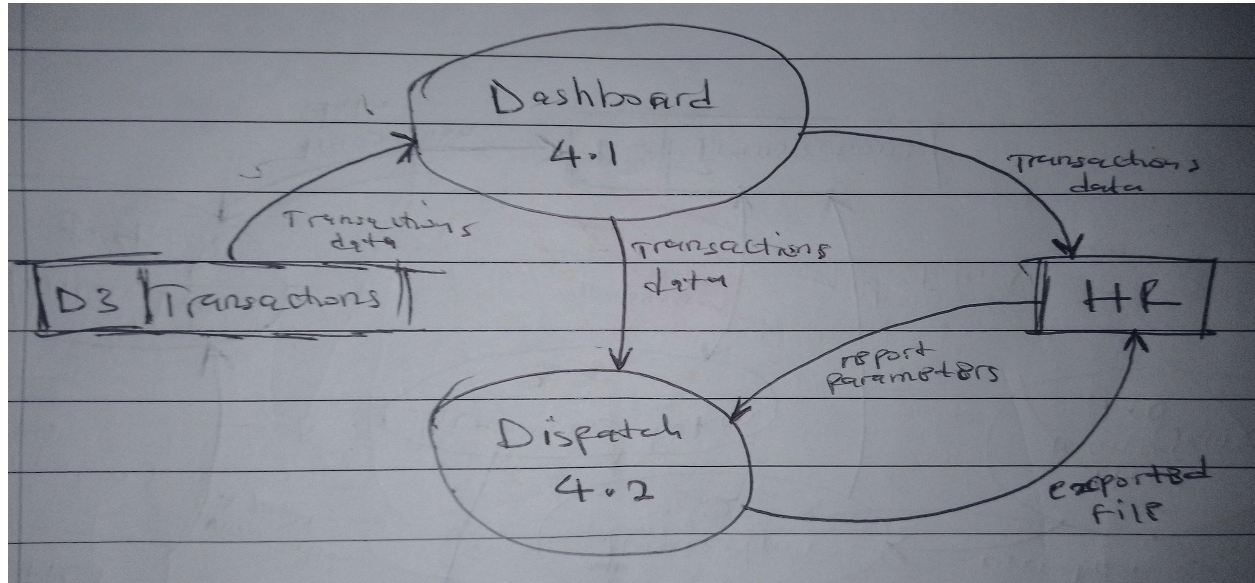
3.0 Order Processing



This is the operational core, managing the transition from a digital order to a physical meal fulfillment.

- **3.1 Fetch Menu:** Receives the current **Menu** from **Process 2.2** and displays it to the **Customer**.
- **3.2 Transaction:** Captures the **Order Selection** from the **Customer**. It uses **User Data** from **Process 1.2** to verify the customer and then logs the complete **Transaction Data** into **D3: Transactions**.
- **3.3 Generate Token:** For every new entry in **D3**, this process creates a unique **Token or QR code**. It sends this back to **D3: Transactions** to update the specific record and then delivers the **Token** to the **Customer**.
- **3.4 Validate:** Receives a **Verification Request** from the **Chef** (via scan) and checks the **Transaction Data** in **D3**. Upon a successful validity check, it updates the record in **D3** to a **Checked Out** status and sends the **Verification Status** (Success/Failure) back to the **Chef**.

4.0 Reporting & Export



This module translates the database records into human-readable administrative reports.

- **4.1 Dashboard:** Periodically reads **Transaction Data** from **D3** to provide a live feed of activity to the **HR Manager**.
- **4.2 Dispatch:** Receives **Report Parameters** (date ranges/formats) from **HR** and pulls the required **Transaction Data** from the dashboard or store to produce the **Exported File**.

Critical Flow	Correction Based Logic
Token Integrity	Tokens are generated <i>after</i> a record exists in D3 and are written back to that record before the user receives them.
Fulfillment Loop	Process 3.4 acts as the final write to D3 , marking orders as Checked Out to prevent double-claiming of tokens.
Verification Gate	Registration (1.1) is purely for logging; verification only happens through the 1.3/1.4 flows controlled by HR.
Pre-Order Sync	Process 2.2 serves as the active bridge between the kitchen's database (D2) and the customer's interface (3.1).